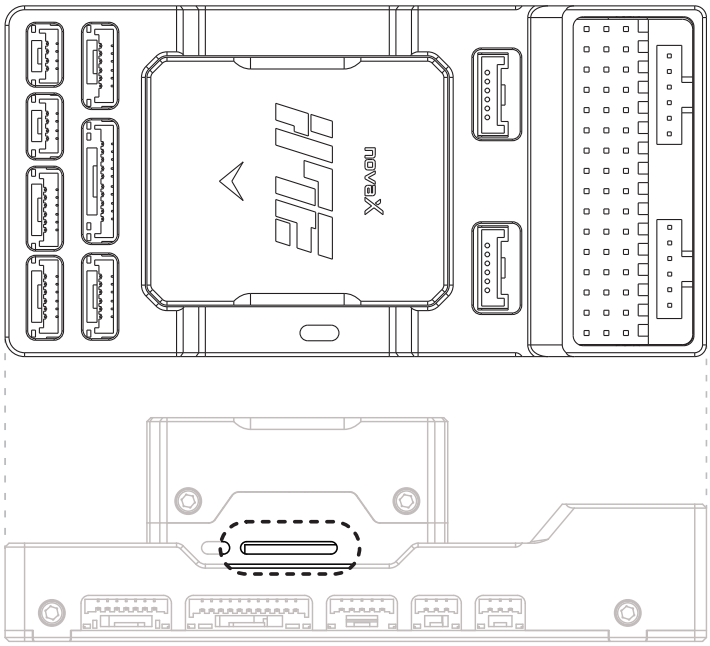
User Guide



novaX

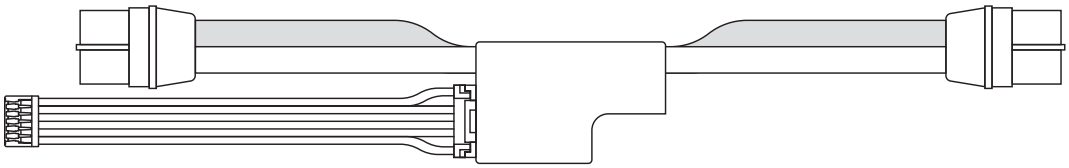
1. Parts List

· Main & Board

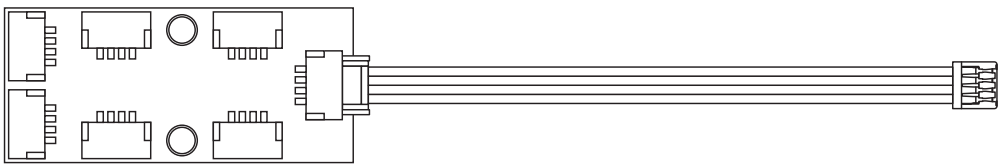


H7E Autopilot

TF Memory Card
(Inserted)



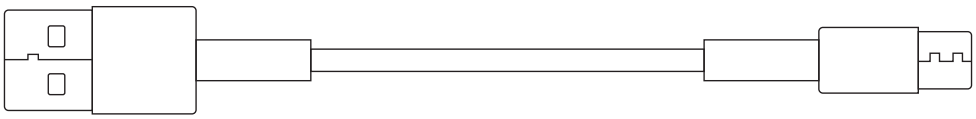
I2C PMU Module * 1



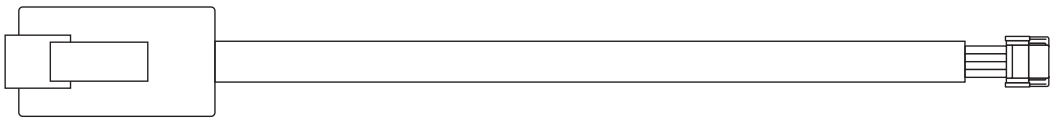
CAN / I2C Extension Board * 1

1. Parts List

· Cable



Type-C Cable



Ethernet Cable

Pin cable type

PPM RC / SBUS OUT Cable * 1	3 - 3 pin
DSM RC Cable * 1	5 - 3 pin
RSSI Cable * 1	5 - 3 pin
SBUS RC Cable * 1	5 - 3 pin
TELEM Cable * 1	6 - 6 pin
CAN Cable * 1	4 - 4 pin
UART4 → I2C Cable * 1	7 - 4 pin
UART4 Cable * 1	7 - 7 pin
SPI6 Cable * 1	11 - 11 pin
Debug Cable * 1	10 - 10 pin
ADIO Cable * 1	8 - 8 pin
GPS Cable * 1	10 - 6 pin

2. Specifications

· Hardware Specifications (1)

MCU	STM32H753, ARM Cortex-M7, 480 MHz
RAM / Flash	1 MB / 2 MB
IMU	BMI088 (Accel/Gyro)
Secondary IMU	ICM-42688-P, ICM-20649
Magnetometer	RM3100
Barometer	2× ICP-20100
Operating Voltage	4.75 – 5.7 V (Rated 5 V)
USB Input	4.75 – 5.25 V
Servo Rail	0 – 9.9 V
PWM Output	16 CH
RC Input	S.Bus, PPM, DSM / Spektrum
RSSI Input	Analog / PWM
UART	7 Port
I2C	3 Port
CAN	2 Port
ADC	VBat/Current + Aux Analog Input
Size	45 × 90 × 29.2 mm
Weight	99 g (Core 43g + Baseboard 56g)

2. Specifications

· Hardware Specifications (2)

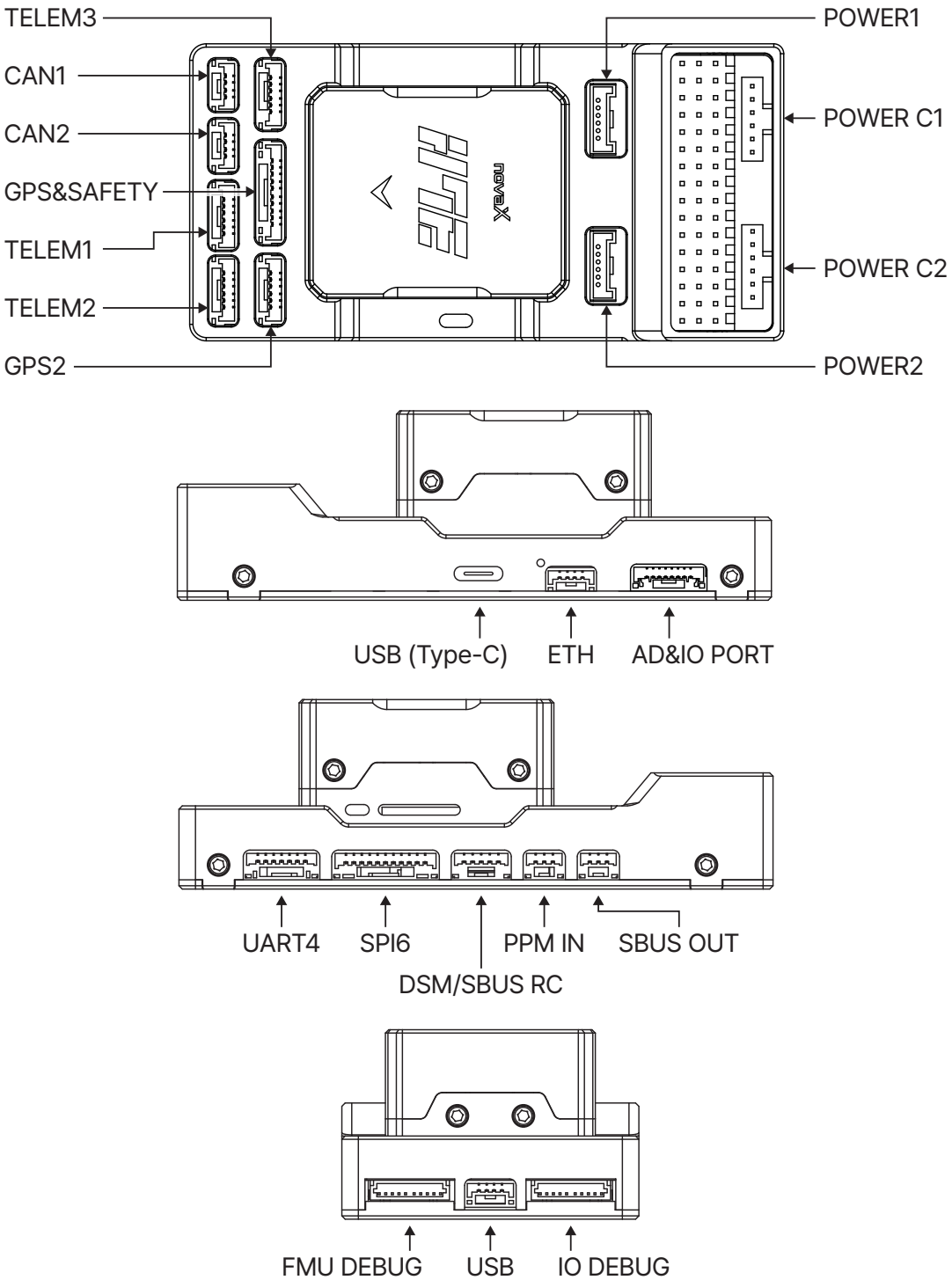
Mount Hole	Pixhawk FMUv6x Standard
Operating Temp	-20 ~ +85 °C
Supported F/W	ArduPilot
Ethernet	100 Mbps x 1 Port

· Interfaces

PWM Outputs	16	POWER	2
GPS Interface	2	TELEM (Telemetry)	3
CAN	2	PPM RC	1
SBUS/DSM/RSSI	1	SBUS OUT	1
FMU DEBUG	1	IO DEBUG	1
ETH (Ethernet)	1	SPI6	1
AD&IO PORT	1	UART4	1
USB Interface	2	TF Card Slot	1
Power Module	I2C PMU (INA2xx) POWER 1 · 2 port		

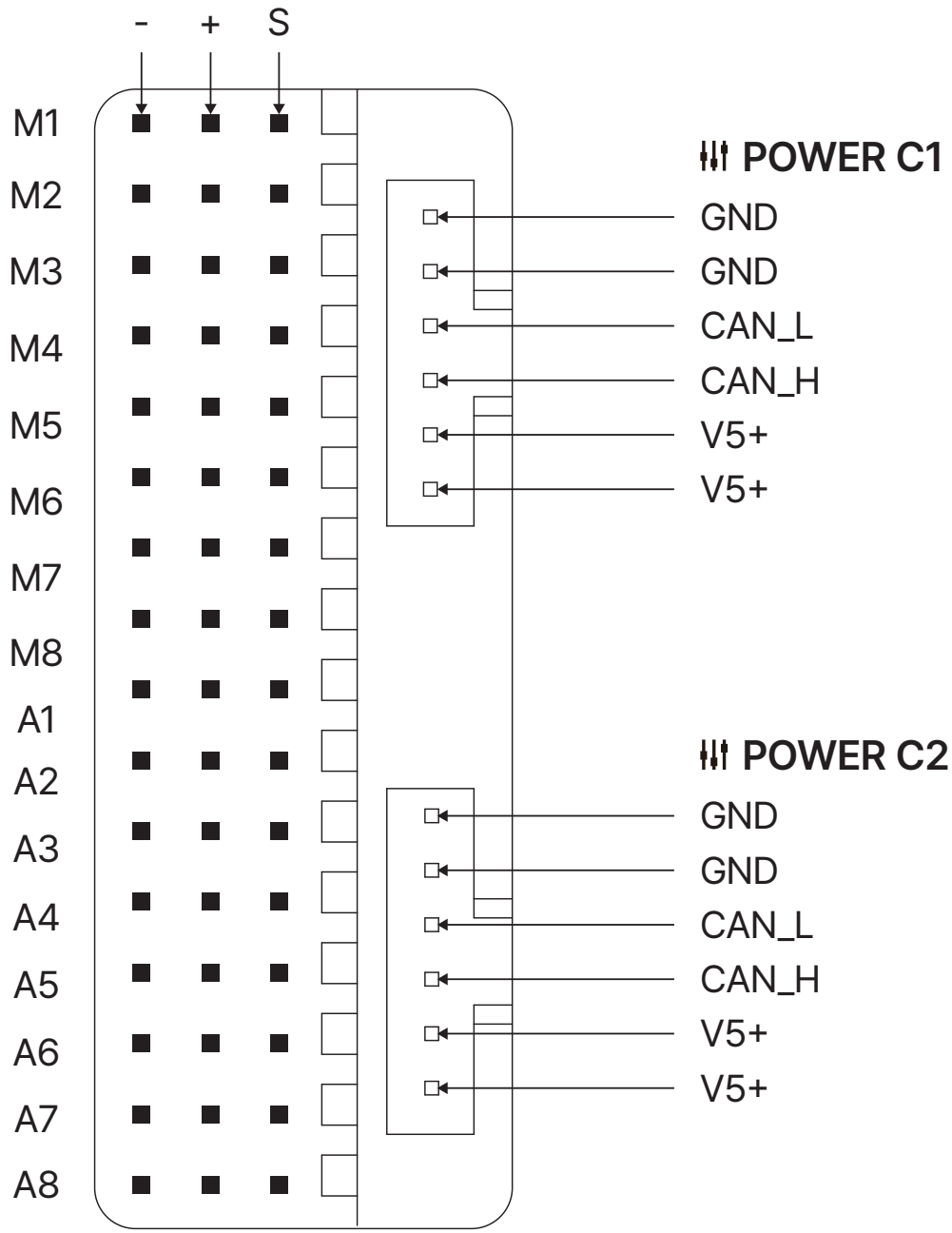
3. Pinouts

· Pinmap



3. Pinouts

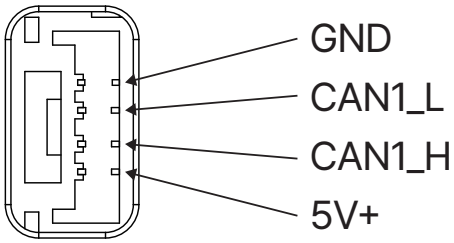
· Top (1)



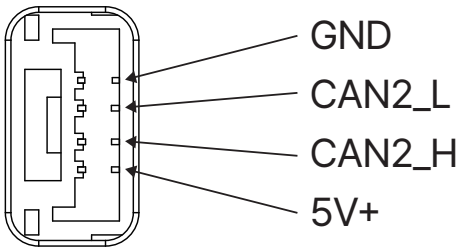
3. Pinouts

· Top (2)

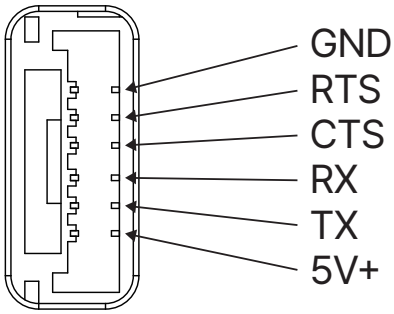
CAN1



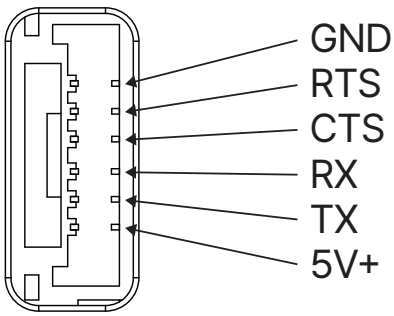
CAN2



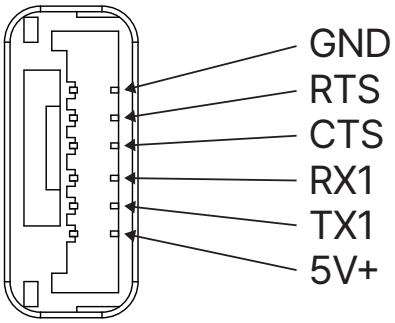
TELEM1



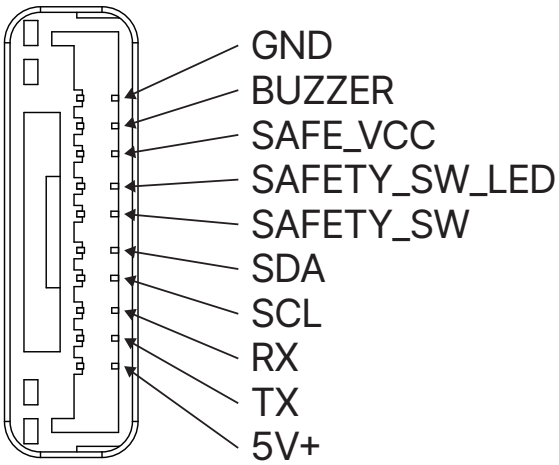
TELEM2



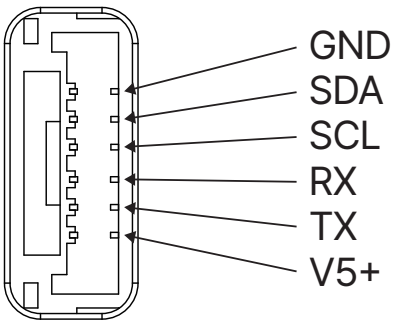
TELEM3



GPS&SAFETY



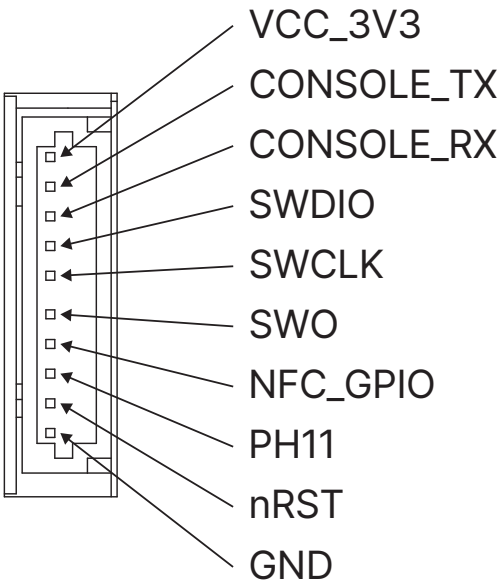
GPS2



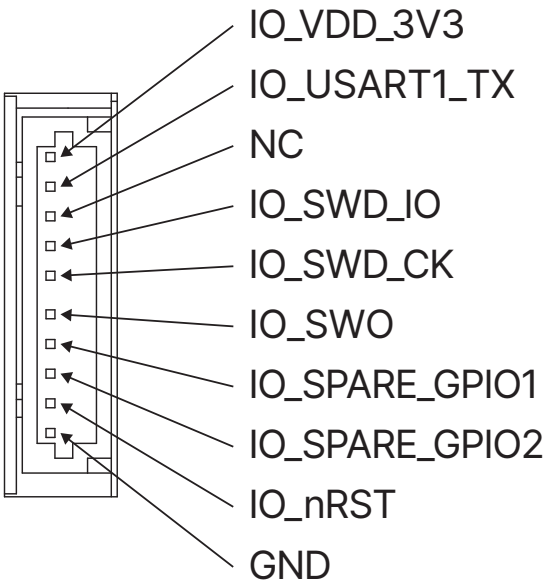
3. Pinouts

· Front

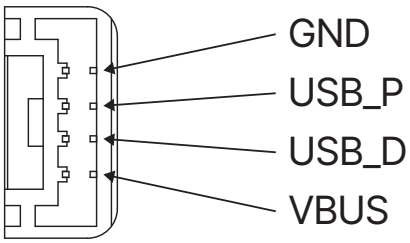
FMU DEBUG



IO DEBUG



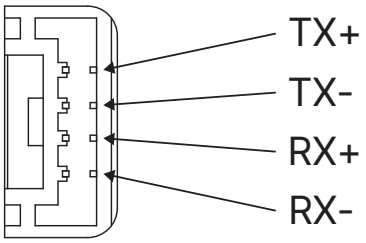
USB



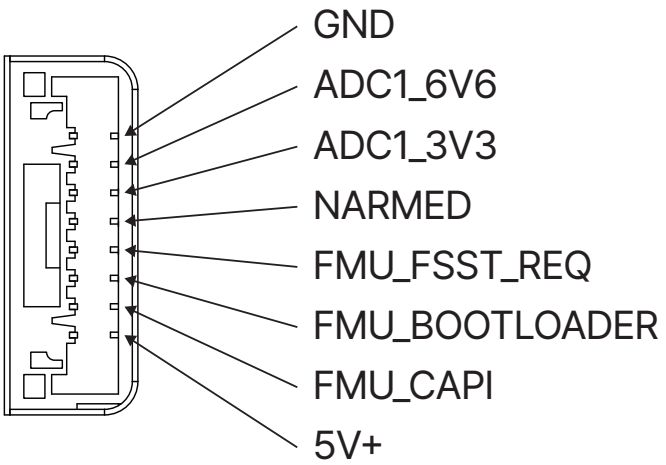
3. Pinouts

· Left

ETH



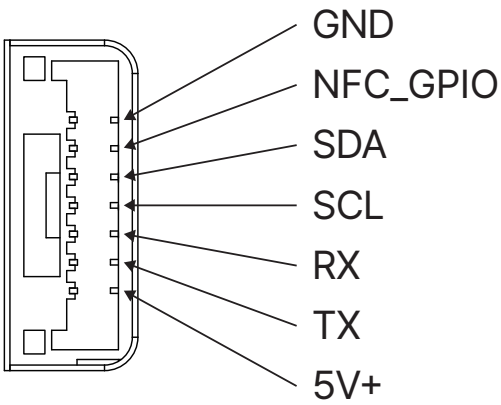
AD&IO PORT



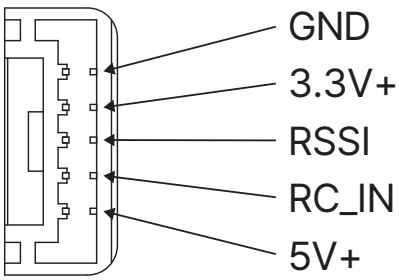
3. Pinouts

· Right

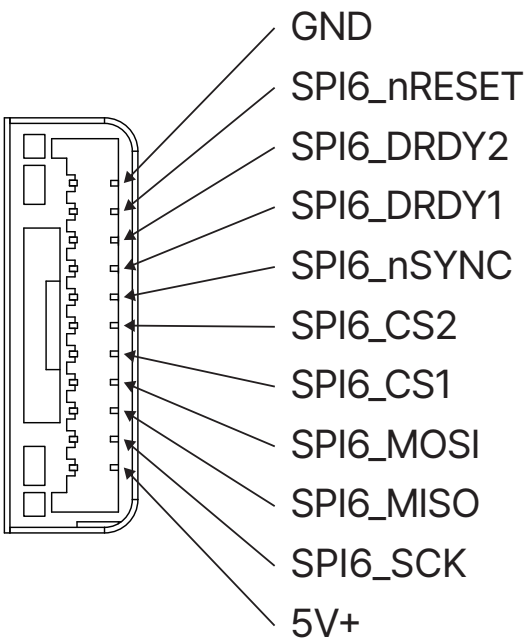
UART4



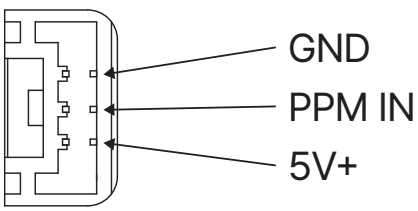
DSM/SBUS RC



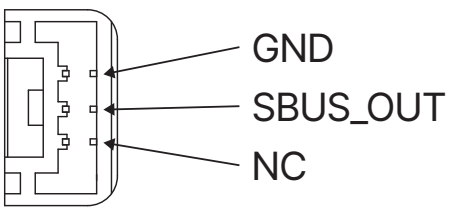
SPI6



PPM IN



SBUS OUT




novaX
HTE